Page Replacement Algorithms

- Page replacement algorithms are the procedures utilizing which an Operating System chooses which memory pages to swap out, write to disk when a page of memory should be allocated.
- Paging happens at whatever point a page fault happens and a free page can't be utilized for allocation reason accounting to reason that pages are not available or the quantity of free pages is lower than required pages.
- At the point when the page that was chosen for replacement and was paged out, is referenced once more, it needs to read in from disk, and this requires for I/O completion.
- This process decides the nature of the page replacement algorithm: the lesser the time waiting for page-ins, the better is the algorithm.
- A page replacement algorithm takes a gander at the restricted data about getting to the pages given by hardware, and attempts to choose which pages ought to be replaced to limit the complete number of page misses, while offsetting it with the expenses of primary storage and processor time of the algorithm itself.
- There are a wide range of page replacement algorithms. We assess a algorithms by running it on a specific string of memory reference and registering the quantity of page issues,
- There are two main parts of virtual memory, Frame allocation and Page Replacement. It is imperative to have the ideal frame allocation and page replacement algorithm. Frame allocation is about what number of frames are to be designated to the process while the page replacement is tied in with deciding the page number which should be supplanted so as to make space for the mentioned page.

What if the algorithm is not optimal?

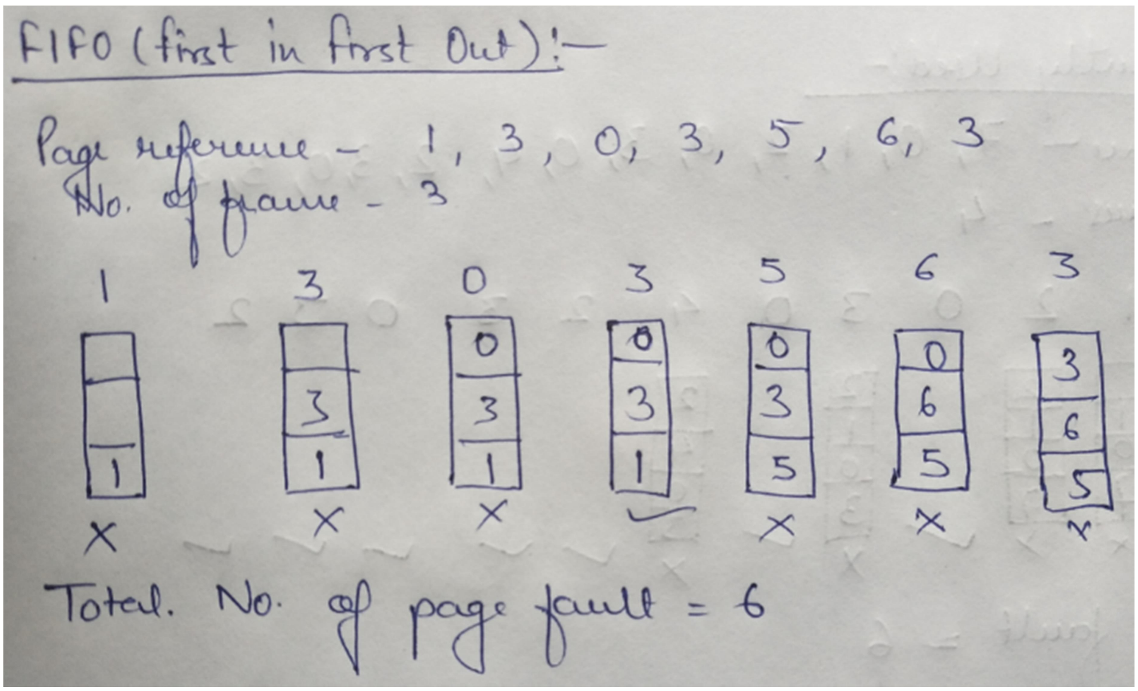
1. In the event that the quantity of frames which are dispensed to a process isn't adequate or exact at that point there can be an issue of thrashing. Because of the absence of frames, a large portion of the pages will reside in the main memory and consequently more page faults will happen.
   Nonetheless, on the off chance that OS allots more frames to the process, at that point there can be internal discontinuity.
2. On the off chance that the page replacement orithmalgorithm isn't ideal, at that point there will likewise be the issue of thrashing. On the off chance that the quantity of pages that are supplanted by the mentioned pages will be referred sooner rather than later at that point there will be progressively number of swap-in and swap-out and along these lines the OS needs to perform more replacements then common which causes execution inadequacy.
3. In this manner, the assignment of an ideal page replacement calculation is to pick the page which can restrict the thrashing.

Types of Page Replacement Algorithm

- There are different page replacement algorithms. Every algorithm has an alternate strategy by which the pages can be replaced.

First in First out (FIFO) –

- This is the least difficult page replacement algorithm. In this algorithm, the operating system monitors all pages in the memory in a queue, the most oldest page is in the front of the line. At the point when a page should be replaced page in the front of the queue is chosen for removal.
- Example Consider page reference string 1, 3, 0, 3, 5, 6 with 3 page frames. Find number of page faults.

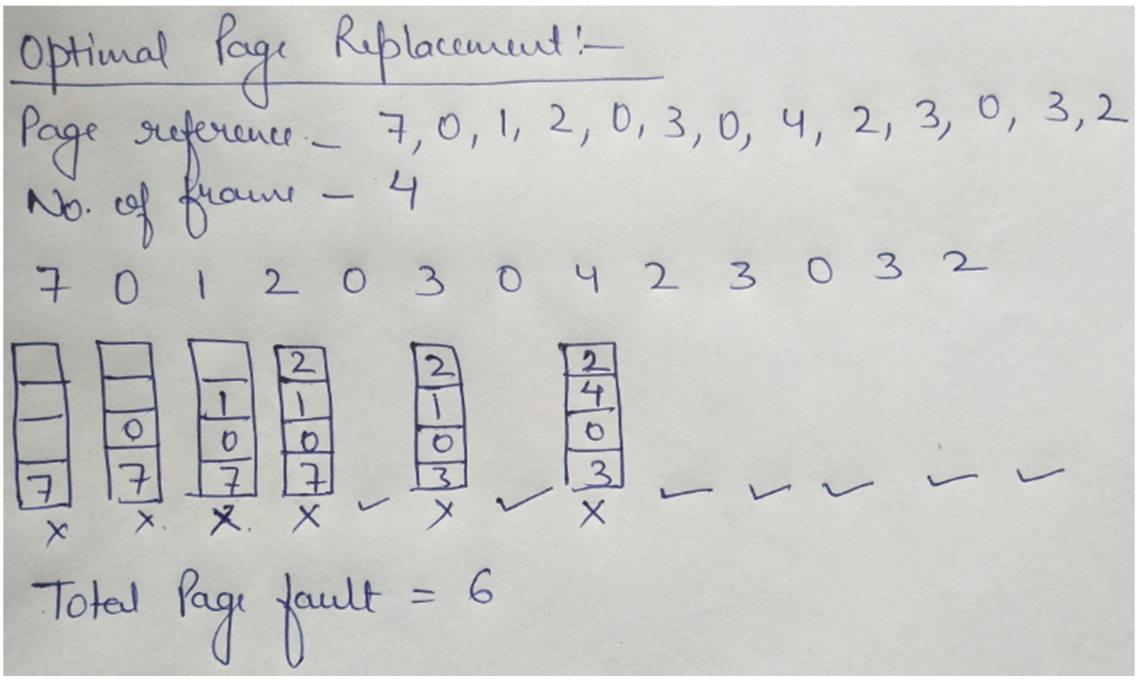


- At first all slots are vacant, so when 1, 3, 0 came they are distributed to the unfilled spaces — > 3 Page Faults.
- At the point when 3 comes, it is in memory so — > 0 Page Faults.
- At that point 5 comes, it isn't available in memory so it replaces the most oldest page space i.e 1. — >1 Page Fault.
- 6 comes, it is likewise not available in memory so it replaces the most established page space i.e 3 — >1 Page Fault.
- At long last when 3 come it isn't avilable so it replaces 0 — > 1 page fault

- **Belady's anomaly** – Belady's anomaly demonstrates that it is conceivable to have more page faults when expanding the quantity of page frames while utilizing the First in First Out (FIFO) page replacement algorithm. For instance, on the off chance that we consider reference string 3, 2, 1, 0, 3, 2, 4, 3, 2, 1, 0, 4 and 3 slots, we get 9 all out page faults, however on the off chance that we increment spaces to 4, we get 10 page faults.

Optimal Page replacement

- This algorithm replaces the page which won't be alluded for such a long time in future. Despite the fact that it cannot be for all intents and purposes implementable however it very well may be utilized as a benchmark. Different algorithms are contrasted with this regarding optimality.
- Example: Consider the page references 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, with 4 page frame. Discover number of page faults.

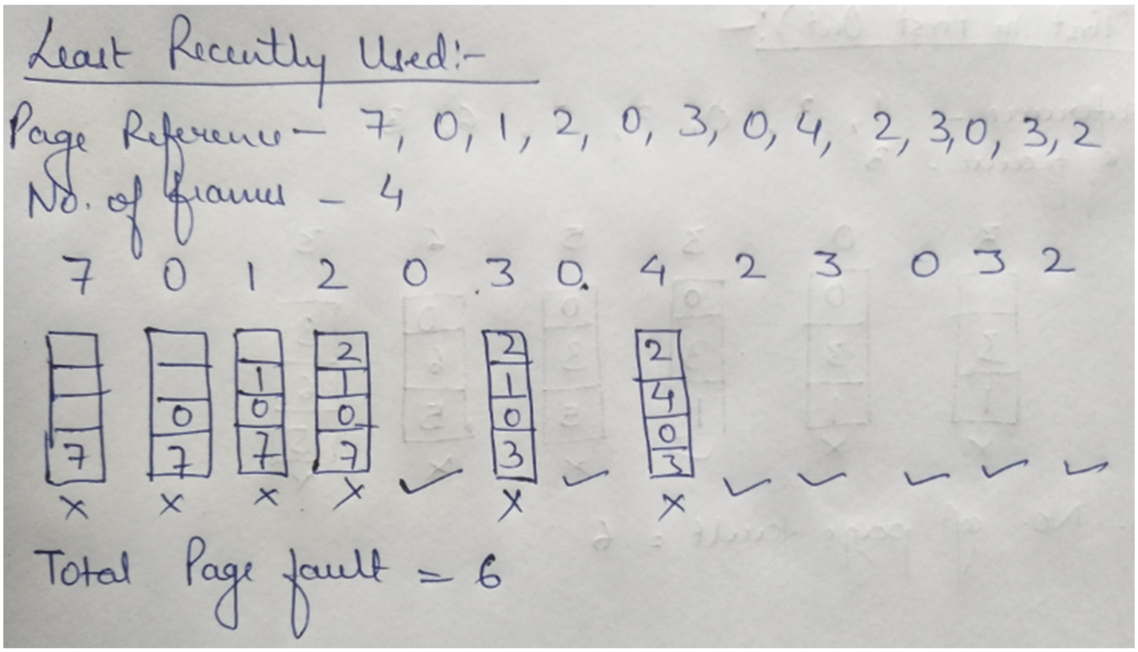


- At first all openings are vacant, so when 7 0 1 2 are assigned to the unfilled spaces —> 4 Page faults
- 0 is as of now there so —> 0 Page fault.
- At the point when 3 came it will replace 7 since it isn't utilized for the longest length of time in future—>1 Page fault.
- 0 is as of now there so —> 0 Page fault.
- 4 will replace of 1 —> 1 Page Fault.

- Presently for the further page reference string —> 0 Page fault since they are as of now accessible in the memory.
- Optimal page replacement is flawless, yet unrealistic by and by as the operating system can't know future requests. The utilization of Optimal Page replacement is to set up a benchmark with the goal that other replacement calculations can be examined against it.

Least Recently Used

- This algorithm replaces the page which has not been alluded for quite a while. This algorithm is only inverse to the optimal page replacement algorithm. In this, we take a look at the past as opposed to gazing at future.
- Example- Consider the page reference string 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2 with 4 page frames. Find number of page faults.



- At first all openings are vacant, so when 7 0 1 2 are apportioned to the vacant spaces —> 4 Page fault
- 0 is now their so —> 0 Page fault.
- at the point when 3 came it will replace 7 since it is least as of late utilized —>1 Page fault.
- 0 is now in memory so —> 0 Page fault.
- 4 will happens of 1 —> 1 Page Fault
- Presently for the further page reference string —> 0 Page fault since they are as of now accessible in the memory.